“Resource Plundering” or Market and Efficiency Driven?

Motivations of China's Direct Investment in Africa

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Abstract

China's Belt and Road Initiative has boosted trade and investment between China and Africa, causing Western unease and doubts of China's investment behavior in the region. Using a sample of 44 African countries that received Chinese investment between 2005 and 2020, this paper employs a system GMM model to examine the effects of host-country natural resource endowments, market size, and efficiency factors on China's investment in Africa. It also divides African countries into low-income economies and middle-income-and-above economies to further analyze China's investment motives across countries at different income levels. The results confirm the paper's two hypotheses: China's investment in Africa is jointly driven by host-country natural resources, market size, and efficiency factors; the effect of natural resources on China's investment in Africa is limited and becomes progressively weaker as the economic level of African host countries rises. Resource-seeking is not the only motive behind China's direct investment in Africa, and there is no "resource plundering"; market-seeking and efficiency-seeking are the decisive factors.

Keywords. resource plundering; market size; efficiency-seeking; China's direct investment in Africa

1. Introduction

According to statistics from the Ministry of Commerce, mainland China's stock of direct investment in Africa increased from USD 490 million in 2003 to USD 43.39 billion in 2020, with an average annual growth rate as high as 30.18 percent. In recent years, African countries have responded actively to China's Belt and Road Initiative, and China's investment in Africa has grown rapidly, heightening Western unease and doubts about Chinese investment behavior in Africa. At the same time, China's investment in Africa has increasingly been driven by the market size and efficiency conditions of African host countries. Data from China's Ministry of Commerce show that between 2013 and 2020 the sectoral composition of mainland China's investment in Africa changed, with more and more investment shifting from mining to construction and manufacturing. On the one hand, the shares of manufacturing and infrastructure investment in China's direct investment in Africa increased steadily. In 2018, China accounted for more than 30 percent of Africa's infrastructure projects. On the other hand, Chinese investors increasingly value the development potential represented by the market size and efficiency of African host countries, while the attraction of resource endowments has been shrinking as the economic level of African host countries rises. Between 2013 and 2020, the share of mainland China's direct investment stock in African mining fell by 5.8 percentage points and was gradually overtaken by manufacturing.

What, then, motivates China's investment in Africa? Some scholars argue that the main motive of China's investment in Africa is the appropriation of host-country natural resources (Gold *et al.*, 2019); others argue that the natural resources of African host countries are not an important factor attracting Chinese investment (Chen *et al.*, 2018); still other studies find that natural resources do play a role in attracting Chinese investment, but that market, efficiency, infrastructure, institutions, and other factors are more important (Chen *et al.*, 2012). This controversy may stem from three causes. First, there are relatively few analyses of the motives behind China's direct investment in Africa. Second, most existing studies examine only a little more than 20 African countries, so their sample data are limited. Third, current research seldom considers differences in China's investment motives across different types of African countries. Further evidence is therefore needed on the motives of China's investment in Africa.

2. Literature Review and Theoretical Hypotheses

The theory of multinational investment motives originated in Hymer's theory of monopoly advantage, which holds that multinational investment arises because multinational corporations possess monopoly advantages over host-country firms in capital, technology,

information, and the like (Hymer, 1960). Yet this theory cannot explain investment behavior by developing countries when they do not possess relative monopoly advantages over developed countries, and this limitation prompted deeper academic exploration. Vernon's product life cycle theory argues that when products enter the mature and standardized stages of their life cycle, firms engage in international investment to reduce costs (Vernon, 1966). Buckley and Casson proposed internalization theory, which holds that multinational investment arises when firms substitute internal markets for external markets to reduce transaction costs (Buckley & Casson, 1976). Building on the studies above, Dunning proposed the eclectic theory of international production, arguing that ownership advantages, internalization advantages, and location advantages jointly determine multinational investment behavior (Dunning, 1977). Kojima proposed the theory of marginal industrial expansion, arguing that multinational investment occurs when firms transfer abroad those production activities in which the home country is at a comparative disadvantage, thereby maintaining the durability of its comparative advantages (Kojima, 1978). In 1981, Dunning further refined the eclectic theory of international production, laying the theoretical foundation for the motives of multinational investment (Dunning, 1981). This paper follows Dunning's eclectic theory of international production to examine the motives behind China's investment in Africa.

According to the eclectic theory of international production, the motives for multinational investment can be divided into four types: natural resource-seeking, market-seeking, efficiency-seeking, and strategic asset-seeking. Natural resource-seeking refers to outward investment undertaken by firms to obtain higher-quality and lower-cost natural resources when certain raw materials or energy sources are scarce or costly. Market-seeking refers to outward investment undertaken to enter host-country or third-country markets. Efficiency-seeking refers to overseas investment aimed at raising input efficiency or realizing economies of scale by taking advantage of differences in factor endowments, institutional arrangements, and demand patterns. Strategic asset-seeking refers to outward investment in search of brands, marketing channels, technology, and other strategic assets to achieve long-term strategic goals. Because Africa is generally economically underdeveloped and, owing to its own constraints, does not enjoy advantages in strategic assets, strategic asset-seeking is not discussed in this paper.

Natural resource endowments have an important foundational effect on FDI (foreign direct investment). At the early stage of economic underdevelopment, when market size, infrastructure, and high-quality labor are lacking, host countries often attract foreign capital by developing and exporting natural resources. In addition, countries endowed with abundant

mineral and energy resources are usually better able to attract foreign investment. However, as host-country economic levels rise, the attractiveness of resource endowments to foreign investment gradually weakens. The same is true of China's outward investment. Research shows that China's investment in Asian countries is mainly motivated by market-seeking, whereas in high-income Asian countries, the resource-seeking motive is not significant (Kamal *et al.*, 2019). Africa remains economically underdeveloped, and many countries are still at the low-income stage, so attracting foreign capital through natural resource endowment advantages has been an unavoidable choice. But as African countries' economic levels rise, the industrial composition of China's investment in those countries becomes more diversified, and factors such as market size and labor costs play a larger role in attracting Chinese investment. Li *et al.* (2017) argue that because China's investment in Africa has gradually shifted toward manufacturing, construction, and services, the resource-seeking motive was not significant during 2009-2014. Utesch-Xiong & Kambhampati (2022) likewise argue that China's investment in middle-income African countries is market-seeking, while a resource-seeking motive exists only in low-income countries.

Market-seeking has become a key factor in outward investment worldwide and in China as well. When a host country's market is sufficiently large, outward investment aimed at expanding that market can avoid the transportation costs associated with direct exports and can further stimulate investment by enabling firms to better understand the host-country market and to develop goods and services more deeply (Buckley *et al.*, 2007). More and more Chinese construction and manufacturing firms view Africa as a huge market for goods and services and invest there accordingly (Liu *et al.*, 2017). In addition, firms' outward direct investment can also involve a "third-party effect": exporting to third-country markets through the host country to reduce the trade barriers associated with direct exports. Some low-income African countries enjoy preferential trade arrangements such as duty-free and quota-free access to the European Union and the United States, which encourages Chinese firms to invest and produce in those African countries and then export to third countries to lower trade costs effectively. As Sanfilippo (2010) argues, beyond natural resources, China's investment in Africa is also motivated by the policy advantages that low-income African countries enjoy in exporting manufactured goods to third countries.

Efficiency-seeking, which relies on host-country factor advantages to reduce production costs, is increasingly becoming an important factor in outward investment. When the investing home country and the host country are at different levels of economic development, firms use differences in factor availability and costs across countries to pursue efficiency, thereby

concentrating capital- and technology-intensive investment activities in developed countries while shifting labor- and natural-resource-intensive production to developing countries. Because African countries lack capital and face high investment risks, their returns on investment are relatively high, so FDI in Africa has a pronounced efficiency-seeking motive (Okafor *et al.*, 2015). With China's economic development and rising labor costs, Chinese firms have also become increasingly active in searching overseas for low-cost production alternatives. Host-country efficiency factors, resource endowments, and market size all have significant effects on attracting Chinese investment, and low host-country wages significantly attract Chinese capital (Rao & Zhu, 2013). How do labor costs affect FDI? When labor costs rise beyond what multinational enterprises can bear, they will reduce investment in that country and transfer labor-intensive production to countries or regions with lower labor costs. Hou *et al.* (2021) find that in regions with lower labor quality, multinational firms prefer higher wages to ensure higher labor quality; by contrast, in regions with higher labor quality, the preference for lower labor costs means that higher wages restrain growth in investment.

Although African countries have developed rapidly overall, their actual economic levels vary widely because there are many countries and development are uneven across them. Following the World Bank classification, this paper divides African countries into low-income countries and middle-income-and-above countries. Among the 44 African countries in the sample, 23 are low-income countries and 21 are middle-income-and-above countries, the vast majority of which are middle-income countries[1]. According to United Nations data, Africa's total population was about 1.3 billion in 2019. More than 50 percent of the population was aged 15-64, while the share aged 65 and above was only 3.4 percent. Based on Africa's rapid but uneven economic development, large market potential, and relatively low labor costs, this paper proposes two hypotheses based on the above analysis:

Hypothesis 1. China's investment in Africa is driven by a combination of host-country natural resources, market size, and efficiency factors.

Hypothesis 2. The effect of natural resources on China's investment in Africa is limited, and this effect becomes progressively smaller as the economic level of African host countries rises.

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[1] The 23 low-income countries are Benin, Burundi, Chad, Comoros, the Democratic Republic of the Congo, Ethiopia, Gambia, Guinea, Kenya, Liberia, Madagascar, Malawi, Mali, Mauritania, Mozambique, Niger, Rwanda, Sierra Leone, Sudan, Tanzania, Togo, Uganda, and Zimbabwe; the 21 middle-income-and-above economies are Algeria, Angola, Botswana, Cabo Verde, Cameroon, the Republic of the Congo, Cote d'Ivoire, Egypt, Equatorial Guinea, Gabon, Ghana, Lesotho, Mauritius, Morocco, Namibia, Nigeria, Senegal, Seychelles, South Africa, Tunisia, and Zambia.

3. Model Specification and Empirical Analysis

3.1 Model Construction and Data Tests

This paper uses China's direct investment in Africa as the dependent variable and selects host-country natural resources, market size, and efficiency factors as the core explanatory variables. These are measured, respectively, by natural resource rents[2] as a percentage of GDP, total population, and per capita GNI. Drawing on existing research, the control variables are the World Bank's Worldwide Governance Indicators (WGI), the number of mobile cellular subscriptions per 100 people, and education expenditure as a percentage of GNI, which proxy for governance capacity, infrastructure, and education levels in African countries.

To study the motives of China's direct investment in Africa, the paper employs a dynamic panel model. The direct investment in Africa discussed here refers only to mainland China's direct investment in Africa. The specific model is:

$$\ln OFDI_{it} = \alpha_0 + \alpha_1 lnOFDI_{it-1} + \alpha_2 resource_{it} + \alpha_3 lnpop_{it} + \alpha_4 lnGNIC_{it} + \alpha_5 WGI_{it} + \alpha_6 lnmobile_{it} + \alpha_7 education_{it} + \varepsilon_{it} \quad (1)$$

In Equation (1), $OFDI_{it}$ represents China's stock of direct investment in Africa; $resource_{it}$ represents host-country natural resource rents as a percentage of GDP; pop_{it} represents the host country's total population; $GNIC_{it}$ represents the host country's per capita gross national income; WGI_{it} represents the host countries worldwide governance indicator value; $mobile_{it}$represents the number of mobile cellular subscriptions per 100 people in the host country; $education_{it}$ represents education expenditure as a percentage of GNI in the host country; α_0 is the constant term; and ε_{it} is a random error term.

Because of data completeness and availability, the paper selects 704 observations for 44 African countries receiving Chinese investment during 2005-2020. Data on China's stock of direct investment in Africa come from the *Statistical Bulletin of China's Outward Foreign Direct Investment*, while all explanatory variables come from the World Bank; some missing data are obtained through interpolation methods. To eliminate the effects of scale and heteroskedasticity, the paper takes logarithms of China's stock of direct investment in Africa and of the explanatory variables' total population, per capita GNI, and mobile cellular subscriptions per 100 people. To determine whether multicollinearity exists among the explanatory variables, the paper tests pairwise correlations. The results show that all pairwise

[2] Natural resource rents include coal, natural gas, oil, mineral, and forest rents.

correlation coefficients are below 0.62. The paper then uses the variance inflation factor (VIF) test and finds that all VIF values are below 2.60 and the average VIF is 1.83, indicating that there is no multicollinearity among the explanatory variables.

3.2 Empirical Results and Robustness Tests

Based on the analysis, the paper estimates a dynamic panel model using system GMM to investigate the motives of China's investment in Africa. System GMM considers both the differenced and level equations and uses lagged first differences as instrumental variables, making it more suitable for short panels and limited samples and allowing it to address the weak-instrument problem. To further test whether the system GMM estimator is valid and reliable, following the suggestions of Bond *et al.* (2001) and Roodman (2009), the paper compares the estimates of the lagged dependent variable from pooled OLS, fixed effects, and difference GMM. If the difference GMM estimate is close to or lower than the fixed-effects estimate, the system GMM estimates are adopted; if the difference GMM estimate is higher than the fixed-effects estimate but lower than the pooled OLS estimate, the difference GMM estimates are used.

Table 1 shows that, for examining the motives of China's investment in Africa, it is appropriate to use a dynamic panel system GMM model. The system GMM estimates indicate a positive relationship between China's stock of direct investment in Africa and its lagged term, significant at the 1 percent level. Natural resource endowments are significantly positively correlated with China's stock of direct investment in Africa, indicating that abundant host-country natural resources are one factor attracting Chinese investment. Market size is significantly positive at the 1 percent level, reflecting China's preference for investing in countries with larger markets, consistent with most of the literature. Per capita GNI, the variable representing efficiency factors, is significantly positive. This finding is consistent with the results of Feng *et al.* (2011) for China. The effect of host-country governance capacity is significantly positive, indicating that China's investment in African countries is not risk-seeking. Infrastructure levels have a significant negative effect on China's investment in Africa; one possible explanation is that Chinese investment seeks to help fill Africa's "infrastructure gap." The education variable is significantly negative; one possible explanation is that in an open factor market, the more highly educated people there are, the more mobile human capital may become across regions (International Organization for Migration, 2020). These results support the first hypothesis advanced in this paper: China's investment in Africa is driven by a combination of host-country natural resources, market size, and efficiency factors.

Table 1. Full-sample regression results from pooled OLS, fixed effects, difference GMM, and system GMM

Variable	(1) Pooled OLS	(2) Fixed effects	(3) Difference GMM	(4) System GMM
$lnOFDI_{it-1}$	0.90*** (0.00)	0.67*** (0.00)	0.66*** (0.00)	0.82*** (0.00)
resource	0.00* (0.08)	0.00 (0.50)	0.01* (0.06)	0.01** (0.02)
lnpop	0.10*** (0.00)	1.36*** (0.00)	1.96*** (0.00)	0.87*** (0.00)
lnGNIC	0.11*** (0.00)	0.39*** (0.00)	0.74*** (0.00)	0.65*** (0.00)
WGI	0.13*** (0.01)	0.26* (0.05)	3.91*** (0.00)	0.99*** (0.00)
lnmobile	-0.12*** (0.00)	-0.00 (0.93)	-0.19** (0.05)	-0.26*** (0.00)
education	-0.03** (0.01)	0.01 (0.82)	0.12** (0.02)	-0.12** (0.03)
Constant	-0.57* (0.06)	-21.28*** (0.00)	-	-14.90*** (0.00)
AR(1)	-	-	0.00	0.00
AR(2)	-	-	0.26	0.75
Hansen test	-	-	0.19	0.19
Observations	660	660	616	660

Note: ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively; figures in parentheses are standard errors. The Arellano-Bond autocorrelation tests and Hansen tests report p-values. The same below.

To further verify that the effect of natural resources on China's investment in Africa is mainly related to the host country's level of development, the paper divides African host countries into low-income and middle-income-and-above groups. Using the same method for empirical analysis, the results again indicate that system GMM estimation is feasible. Because of space limitations, only the system GMM estimates are reported below. Table 2 shows that in both low-income countries and middle-income-and-above countries, China's stock of direct investment in Africa is significantly positively correlated with its first lag, total population, and per capita GNI, and significantly negatively correlated with host-country infrastructure and education levels. However, natural resource endowments are significantly positive only in low-income countries and not significant in middle-income-and-above countries; the governance indicator is not significant in low-income countries but significantly positive in middle-income-and-above countries. This supports the second hypothesis advanced in this paper: the effect of natural resources on China's investment in Africa is limited and gradually declines as

African host countries' economic levels rise, further confirming that the claim that China engages in "resource plundering" in Africa is inconsistent with reality.

Table 2. System GMM regression results in different income groups

Variable	(1) Low-income countries	(2) Middle-income and above countries
$lnOFDI_{it-1}$	0.86*** (0.00)	0.87*** (0.00)
resource	0.01*** (0.00)	0.00 (0.12)
lnpop	0.19*** (0.00)	0.15*** (0.00)
lnGNIC	0.25*** (0.00)	0.25*** (0.00)
WGI	0.05 (0.47)	0.28*** (0.00)
lnmobile	-0.09*** (0.01)	-0.15*** (0.00)
education	-0.03** (0.01)	-0.05*** (0.00)
Constant	-2.89*** (0.00)	-1.93 (0.10)
AR(1)	0.00	0.00
AR(2)	0.72	0.88
Hansen test	0.15	0.47
Observations	345	315

Table 3. Robustness test results

Variable	(1) Full sample	(2) Low-income countries	(3) Middle-income and above countries
$lnOFDI_{it-1}$	0.84*** (0.00)	0.88*** (0.00)	0.87*** (0.00)
mineral	0.02** (0.05)	0.01*** (0.00)	0.00 (0.70)
lnpop	0.75*** (0.00)	0.17*** (0.00)	0.18** (0.01)
lnGNIC	0.61*** (0.00)	0.20*** (0.00)	0.28** (0.02)
WGI	0.77*** (0.01)	0.01 (0.83)	0.36** (0.03)
lnmobile	-0.32*** (0.00)	-0.11*** (0.00)	-0.19*** (0.00)
education	-0.12** (0.04)	-0.03** (0.03)	-0.07** (0.01)
Constant	-12.63*** (0.00)	-2.25*** (0.01)	-2.41 (0.18)
AR(1)	0.00	0.00	0.00
AR(2)	0.58	0.68	0.82
Hansen test	0.16	0.13	0.88
Observations	660	345	315

Note: In this table, mineral denotes host-country mineral resource rents as a percentage of GDP.

On this basis, the paper uses mineral resources instead of natural resources as the proxy variable for resource endowment to conduct a robustness test, and the results are reported in Table 3. The relevant conclusions are consistent with those above and further verify the paper's two hypotheses.

4. Conclusions and Recommendations

China's direct investment in Africa exhibits a certain resource-seeking motive, but resource-seeking is not its only motive, and there is certainly no "resource plundering." Moreover, the resource-seeking motive is mainly related to the host country's level of development and becomes progressively weaker as that level rises. Market-seeking and efficiency-seeking are the decisive drivers of China's direct investment in Africa. The larger the host-country market, the more conducive it is to attracting Chinese investment. In addition, because labor costs in host countries are overall at a low level, a moderate rise in wages can better enhance labor productivity and worker incentives, thereby creating more profits and further attracting Chinese investment. Host-country infrastructure and education levels have significant negative effects on China's direct investment in Africa, whereas the global governance indicator is insignificant only in low-income countries; African countries with stable political situations and relatively sound government governance are more attractive to Chinese investors. Accordingly, China should be guided by the goal of building a China-Africa community with a shared future in the new era, accelerate implementation of the "going global" strategy, and encourage and support Chinese enterprises to deepen both the breadth and depth of investment in Africa. Chinese enterprises should be encouraged and supported to make full use of African countries' large market potential, low labor costs, and host-country advantages in exporting to third-country markets, so that China's investment in Africa can more extensively promote the development of advanced manufacturing, the upgrading of services, and agricultural prosperity in Africa. At the same time, the identification and assessment of investment risks in African countries should be strengthened, and Chinese enterprises' awareness of guarding against political risks in Africa should be enhanced so as to respond properly to the effects on Chinese investment caused by factors such as frequent changes of regime and social unrest in African countries.

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